
Course Name: Machine Learning for Earth System Sciences

Week 03 - Assignment 03 - (Jul-2026)

TYPE OF QUESTIONS: MCQ/MSQ

Number of questions: 10

Total marks: 10x1= 10

QUESTION 1

Following a powerful cyclone, disaster management authorities need to automatically locate damaged buildings in high-resolution satellite images. The AI system should identify each building and predict its location using bounding boxes so that rescue teams can rapidly assess the extent of damage. Which of the following approaches is most appropriate for this task?

- A. Image Classification (e.g., ResNet)
- B. Object Detection (e.g., YOLO)
- C. Image Segmentation (e.g., U-Net)
- D. Regression (e.g., Linear Regression)

Answer: B

Detailed Solution:

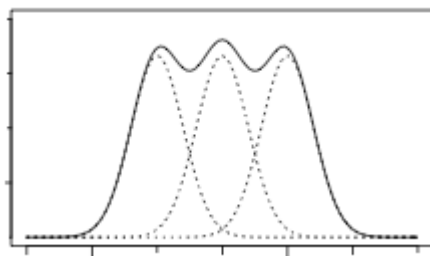
Object detection models (like YOLO, Faster R-CNN, or SSD) are explicitly built to predict both class labels (e.g., "damaged building") and the exact spatial coordinates of a bounding box surrounding every single instance of that class.

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QUESTION 2

After analysing the marks of all students, Prof. Kumar has decided to form probabilistic clusters for the 400 students enrolled in his course. (The given figure shows the distribution of student marks)



Which of the following methods will be suitable for him to accomplish it?

- (A) Bayesian Network
- (B) Gaussian Mixture Model
- (C) K-Means clustering
- (D) Hard Clustering

Answer: B

Detailed Solution:

Gaussian Mixture Model is used for soft clustering (probabilistic clustering).

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QUESTION 3

Let X is a 4×4 matrix with each element $X_{ij} = (i+j)$, where i represents row number and j represents the column number ($0 \leq i \leq 3$ and $0 \leq j \leq 3$). What will be the output matrix Y if we apply max-pooling on X for a block size of 2 and stride equals to 2.

(A)

2	4
4	6

(B)

1	3
4	6

(C)

0	2
2	4

(D)

6	6
6	6

Answer: A

Detailed Solution:

$X =$

0	1	2	3
1	2	3	4
2	3	4	5
3	4	5	6

Given, block size=2 and stride=2

Hence,

$Y =$

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2	4
4	6

QUESTION 4

When modelling long-term Earth system processes, a primary limitation of LSTM networks compared to newer architectures like Transformers is:

- (A) They cannot process daily or hourly observations.
- (B) Their sequential processing limits parallelization and makes it more difficult to capture very long-range temporal dependencies.
- (C) They cannot be trained on multivariate Earth system data.
- (D) They require all input time series to have no missing observations.

Answer: B

Detailed Solution:

LSTMs process sequences sequentially, which hinders parallel computation and makes learning very long-range dependencies (e.g., decadal or centennial climate variability) more challenging than Transformers with self-attention.

QUESTION 5

Which of the following statement is false for a Bayesian Network?

- (A) Bayesian Network is a Directed Acyclic Graph.
- (B) In the Bayesian Network a node can have multiple parents.
- (C) The conditional distribution of the nodes does not depend on parent variables.
- (D) In Bayesian Network each edge has a direction from parent node to child node.

Answer: C

Detailed Solution:

Bayesian Network has following characteristics:

- (i) Bayesian Network is a Directed Acyclic Graph.
- (ii) In the Bayesian Network a node can have multiple parents.

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- (iii) The conditional distribution of each node depends on parent variables.
- (iv) In Bayesian Network each edge has a direction from parent node to child node.

Hence in question except option C all others are true for Bayesian networks.

QUESTION 6

Which of the following is not a characteristic feature of earth system data?

- (A) High Dimensional
- (B) Multi resolution
- (C) Always Categorical
- (D) Imbalanced class

Answer: C

Detailed Solution:

The characteristics features of earth system data are high dimensional, multi resolution and imbalanced class. The data can be real-valued or discrete scalar or vector. Categorical data is not a characteristic feature for earth systems.

QUESTION 7

There is a coniferous forest in the northwest region of your study area, so you identify it by enclosing it on the map with a polygon (or with multiple polygons). Another polygon is created to encompass a wheat field, another for urban buildings, and another for water. You continue this process until you have enough features to represent a class, and all classes in your data are identified. Each grouping of features is considered a class, and the polygon that encompasses the class is a training sample. This task is an example of?

- (A) Supervised learning / Regression
- (B) Supervised learning / Classification
- (C) Unsupervised learning / Clustering
- (D) Unsupervised learning / Classification

Answer: B

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Detailed Solution:

Since training samples are used to identify different classes, so it's a supervised classification model.

QUESTION 8

Select the most appropriate matching

Method / Tool	Task
A. LSTM	P. Rainfall forecasting from historical meteorological observations
B. CNN	Q. Flood inundation mapping using satellite imagery and semantic segmentation
C. Auto Encoders	R. Compression of high-dimensional climate model outputs
D. U-Net	S. Land-use/Land-cover classification from satellite imagery

- (A) A - P, B - R, C - S, D - Q
(B) A - P, B - S, C - R, D - Q
(C) A - Q, B - S, C - R, D - P
(D) A - Q, B - S, C - P, D - R

Answer: B

Detailed Solution:

LSTM → Rainfall forecasting (time-series prediction)

CNN → Land-use/Land-cover classification (image-level classification)

Autoencoder → Compression of high-dimensional climate model outputs (dimensionality reduction/representation learning)

U-Net → Flood inundation mapping (pixel-wise semantic segmentation)

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QUESTION 9

Which of the following statements about Boltzmann's machine is/are true?

- I. Unlike a Deep Belief Network, Deep Boltzmann's machines are an entirely undirected model.
- II. Connections between hidden-to-hidden layer are not allowed but between visible-to-visible layer are allowed in a Restricted Boltzmann's machine.
- III. Boltzmann's machines are energy-based models

- (A) I
- (B) I and II
- (C) I and III
- (D) II and III

Answer: C

Detailed Solution:

Restricted Boltzmann's machines are a class of Boltzmann's machine with single hidden layer and with a bipartite connection. (Intra connections between visible and hidden layer nodes are not allowed)

QUESTION 10

Which of the following tasks is least suitable for an LSTM model?

- (A) Soil type classification from a single multispectral satellite image
- (B) Crop yield prediction using a season-long time series of temperature, rainfall, and soil moisture
- (C) Groundwater level prediction using historical groundwater levels and rainfall observations
- (D) Flood forecasting using historical river discharge and rainfall observations



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Answer: A

Detailed Solution:

LSTMs are employable for time series-based tasks that require to remember previous states, Soil type classification based on single multispectral satellite images is the least probable out of the given tasks to use a time series data. Models such as CNNs and ResNet are more appropriate than LSTMs.